

SAIKAT DAS

Backend-Focused Development | DevOps | AI/ML Engineering

Dhaka, Bangladesh | +88 01790070535

saikatdasmoin47@gmail.com | linkedin.com/in/saikatdas47

Portfolio | GitHub



Professional Summary

Computer Science and Engineering graduate focused on backend development, DevOps, and AI/ML engineering. Experienced in developing and deploying web applications using Node.js, Express.js, Django, REST APIs, PostgreSQL, MongoDB, MySQL, Redis, and cloud platforms. Familiar with backend architecture, authentication, caching, database design, deployment workflows, networking fundamentals, and basic DevOps practices. Knowledgeable in ML, deep learning, computer vision, VLMs, LLM concepts, prompt engineering, fine-tuning concepts, and Responsible AI.

Experience

Research Assistant (Internship), ELITE Research Lab LLC Remote; Queens, NY, USA. Areas: Computer Vision, LLM, VLM.	Sep 2026–Present
Back End Developer (Internship), Vyren Remote; Germany.	Sep 2026–Present
Backend AI Engineer – Intern, FlyRank AI Remote.	Sep 2026–Present

Technical Skills

Backend-Focused	DevOps	AI/ML Engineering
Node.js and Express.js; TypeScript and Nest.js; REST API development; PostgreSQL, MongoDB and MySQL; Redis caching; JWT authentication; Email Authentication; Database design and transactions; Load-balancing concepts; System design fundamentals; Scalable backend concepts; API security and optimization.	Git and GitHub; Linux fundamentals; Docker fundamentals; CI/CD fundamentals; AWS fundamentals; Vercel and Render; Firebase and Cloudinary; Environment configuration; Application deployment; Logging and monitoring concepts; Basic server management; Deployment troubleshooting.	Machine Learning; Deep Learning; Computer Vision; Vision-Language Models; Natural Language Processing; Large Language Models; Fine-tuning concepts; Prompt engineering; Responsible AI; Model evaluation; Data preprocessing; PyTorch and TensorFlow.
Programming Languages	JavaScript, TypeScript, Python, C++, SQL, MATLAB, Java, and PHP (basic)	
CS Fundamentals	DSA, Object-Oriented Programming, Operating Systems, Computer Networks, Computer Architecture, Database Management Systems, and Software Engineering	
Development Tools	VS Code, Jupyter Notebook, Postman, GitHub, XAMPP, database management tools	
Problem Solving	Solved 200+ problems on Codeforces, LeetCode, and other online problem-solving platforms	
Self-study Topics	Node/Express routing, middleware, sessions, testing; cloud service models, virtualization, migration, security, SLA management; Agile, CI/CD, build and deployment automation; NumPy, pandas, Matplotlib, data cleaning and visualization.	

Education

B.Sc. in Computer Science and Engineering Ahsanullah University of Science and Technology (AUST), Dhaka CGPA in Last Four Semesters: 3.65/4.00	Graduated 2026 CGPA: 3.32/4.00
Higher Secondary Certificate (HSC) , Chittagong College	2020 GPA: 5.00/5.00
Secondary School Certificate (SSC) , Saint Placid's High School	2018 GPA: 5.00/5.00

Development Projects

CommerceX – AI-Powered Multi-Vendor Marketplace Stack: Node.js, PostgreSQL, Redis, Socket.IO, React, Vite, Docker and Cloudinary Demonstrates complete e-commerce workflows, REST API design, inventory handling, and real-time communication. - Built customer, seller, and administrator workflows with JWT authentication, role-based access, product variants, checkout, order fulfilment, reviews, and shipment tracking. - Used transactional stock updates and Redis caching for reliable inventory, with Socket.IO for real-time admin–seller messaging.	GitHub Live Website
--	-----------------------

Dhaka Tesla Pool – Shared Ride Platform

[GitHub](#) | [Live Website](#)

Stack: React.js, Vite, Node.js, Express.js, MongoDB Atlas, Socket.IO, JWT, Cloudinary and Docker

Demonstrates route-matching algorithms, seat- and distance-based fare calculation, and real-time ride updates.

- Built a ride-sharing system that groups passengers travelling along compatible routes and calculates individual fares from shared distance and booked seats.
- Used Socket.IO for live communication and REST APIs for passenger, ride, authentication, and media workflows.

Stremars – YouTube-Style Video Sharing Platform

[GitHub](#) | [Live Website](#)

Stack: React.js, Vite, Node.js, Express.js, MongoDB, Mongoose, Cloudinary, Multer and JWT

Demonstrates large media-file handling, secure uploads, and the data flow behind a video-sharing platform.

- Implemented video upload, viewing, search, comments, likes, subscriptions, watch history, creator profiles, and community posts.
- Used Multer and Cloudinary to manage videos, thumbnails, avatars, and cover images while storing metadata in MongoDB.

Hotel Management System

- Built user authentication, room management, reservations, payments, inquiries, and administrative features.
- Designed a relational MySQL database with foreign-key constraints and transaction-based booking operations.

AI/ML Engineering Project

Edge-AI Waste Classification – Real-Time Smart Bin

[GitHub](#) | [Demo Video](#)

Stack: Python, TensorFlow Lite, Raspberry Pi, OpenCV, Firebase and Flask

- Developed and evaluated a real-time Edge-AI waste-classification system using computer vision models deployed on Raspberry Pi.
- Integrated image capture, biometric authentication, fill-level monitoring, hardware control, secure logging, and Firebase-based remote monitoring.

Publication Under Review

Saikat Das, et al. “A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging” – under review, submitted April 2026.

Training & Certifications

- **OpenCV University:** Python, PyTorch, TensorFlow-Keras, and Vision-Language Model Bootcamp 2025–2026
- **Google Cloud Skills Boost:** Prompt Design, Responsible AI, Introduction to LLMs, Introduction to Generative AI, and ML Engineer Guide 2025–2026
- **MathWorks:** MATLAB Onramp, Machine Learning Onramp, and Deep Learning Onramp 2026
- **IBM/Coursera:** Introduction to Data Engineering February 2026

Additional Information

Languages	Bengali – Native; English – Intermediate (B1)
Strengths	Backend-focused problem-solving, quick learning, teamwork, technical documentation, adaptability, and attention to detail
Target Roles	Backend Developer, Node.js Developer, Software Engineer, Junior DevOps Engineer, and AI/ML Engineer

References

Ms. Nusrat Jahan

Assistant Professor, Department of CSE
Ahsanullah University of Science and Technology
nusratjahan.cse@aust.edu

Mr. Mustofa Ahmed

Lecturer, Department of CSE
Ahsanullah University of Science and Technology
mustofa.cse@aust.edu

Signature of Saikat Das